F12 4th CAR

Lecture 25: Section 4.1
Angles and Their Measure

Angle - initial side, terminal side, vertex

Standard position of an angle

Positive and negative angles

Coterminal angle

Central angle

Radians

Complementary and suppplementary angles

Degree measure and radian measure

Arc length, s

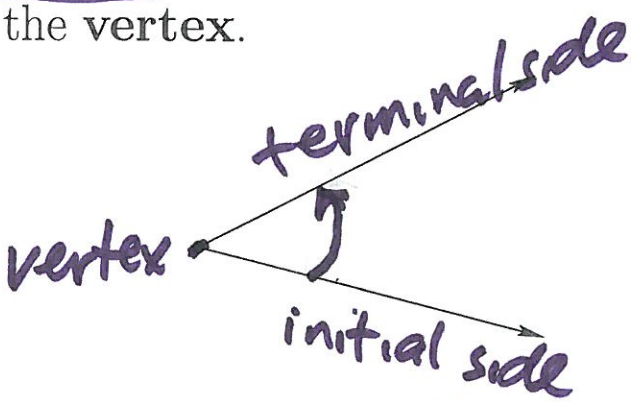
Area of a sector

Linear speed

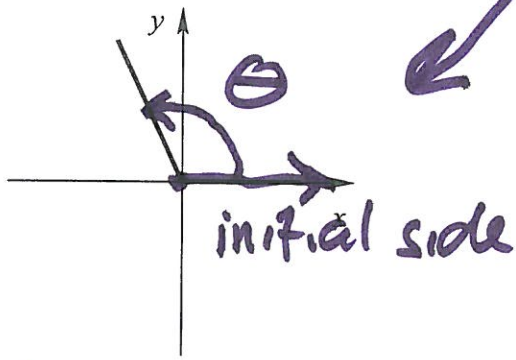
Angular speed

Lecture 25: Section 4.1
Angles and Their Measure

An **angle** is formed by rotating a ray around its endpoint. The starting position of the ray is the **initial side** of the angle, and the position after rotation is the **terminal side** of the angle. The endpoint of the ray is called the **vertex**.

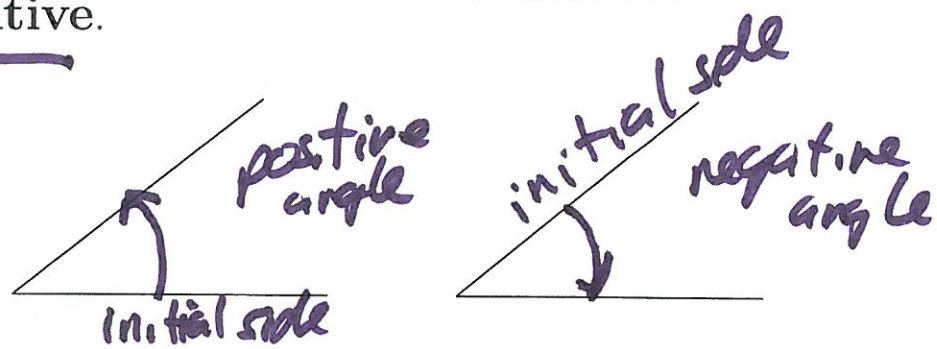


An angle θ is said to be in **standard position** if its vertex is in the origin and its initial side coincides with the positive x-axis.



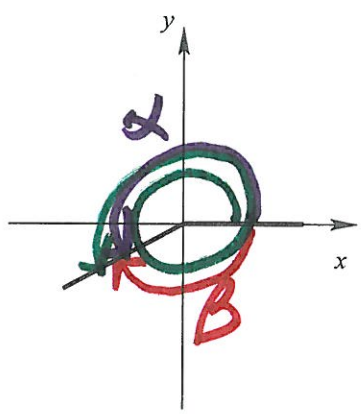
not in standard position

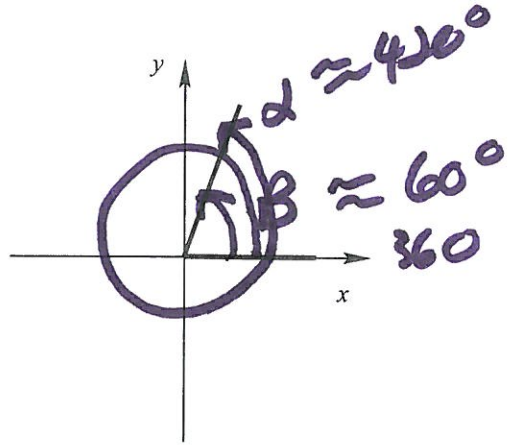
If the rotation is in the counterclockwise direction, the angle is **positive**; if the rotation is clockwise, the angle is **negative**.



Angles α and β are **coterminal angles** if they have the same initial and terminal sides.

210°
−150°





360
210
570°

A **central angle** is an angle whose vertex is at the center of a circle.

Radian Measure

<u>Def.</u> **One radian** is a measure of the central angle that intercepts an arc whose length is equal to the radius. Algebraically, this means that

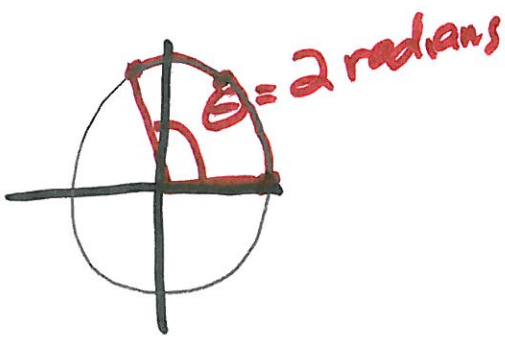
arc length

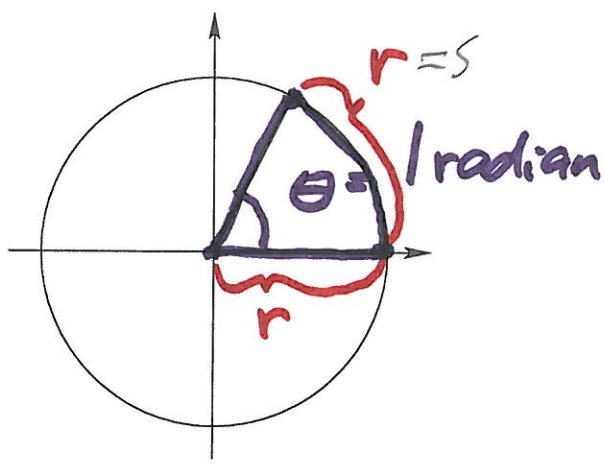
angle in radian measurement

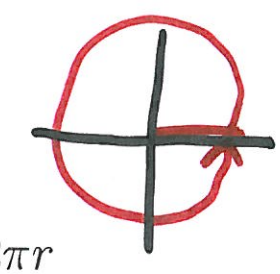
$$\boxed{\theta = \frac{s}{r}}$$

radius

where θ is measured in radians.







NOTE: For the angle $\theta = 1$ revolution:

The length of the arc (circumference) $s = 2\pi r$

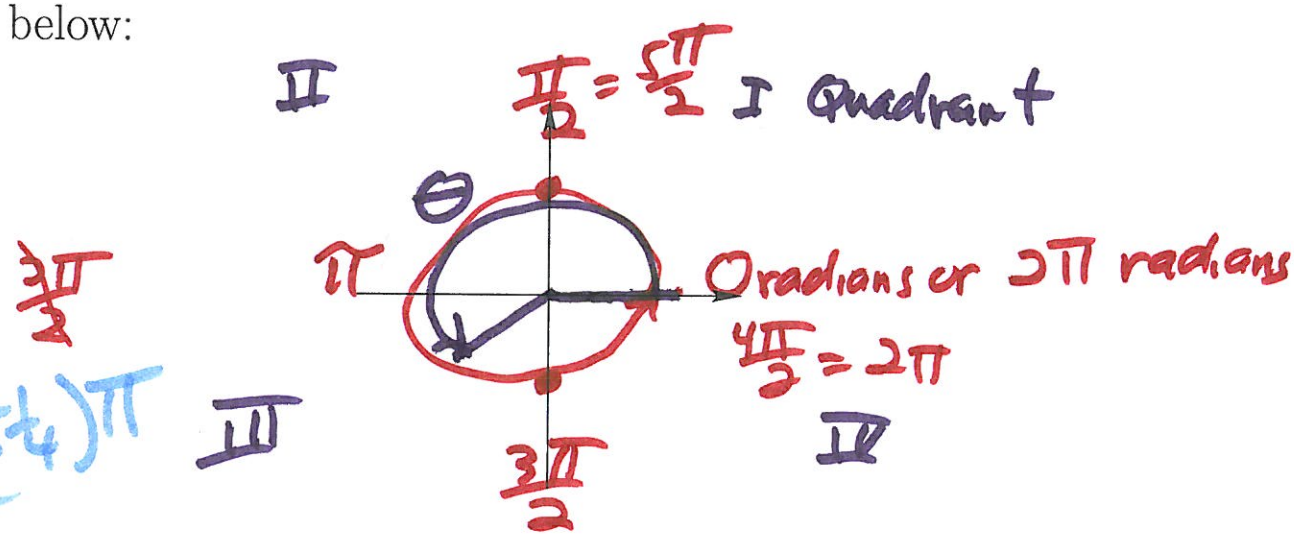
Therefore, $\theta = \dfrac{s}{r} = \dfrac{2\pi \cancel{r}}{\cancel{r}} = 2\pi.$ So 2π is the same as $360°$.

We have,

$$1 \text{ revolution} = 2\pi \text{ radians}$$

circumference of a circle $= \pi d = 2\pi r$

NOTE: If $0 \leq \theta \leq 2\pi$, the standard position of the angle θ in the Cartesian coordinate system is shown below:



NOTE: Given an angle θ, the coterminal angles to θ are

$$\theta + 2n\pi = \theta + (2\pi)n$$

$-\frac{21}{4} = -5\frac{1}{4}$

where n is an integer. $\{\ldots -3, -2, -1, 0, 1, 2, 3 \ldots\}$

ex. Find the angle of [redacted] the smallest positive measure that is coterminal with $\theta = -\dfrac{21\pi}{4}$.

$n=1$ $\quad \frac{-21\pi}{4} + 2\pi(1) = \frac{-21\pi}{4} + \frac{8\pi}{4} = \frac{-13\pi}{4}$

$n=2$ $\quad \frac{-21\pi}{4} + 2\pi(2) = \frac{-21\pi}{4} + \frac{16\pi}{4} = \frac{-5\pi}{4}$

$n=3$ $\quad \frac{-21\pi}{4} + 2\pi(3) = \frac{-21\pi}{4} + \frac{24\pi}{4} = \boxed{\frac{3\pi}{4}}$

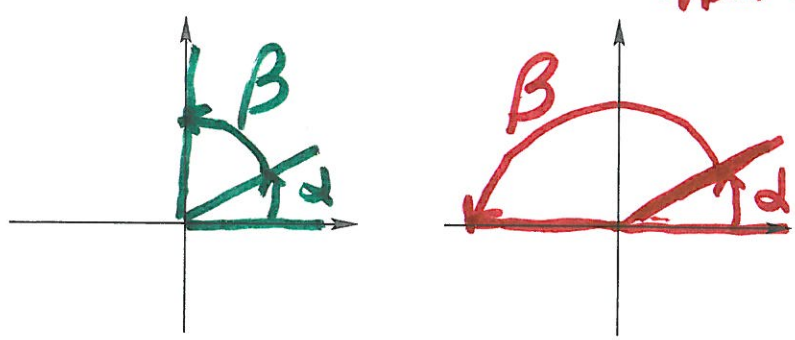
Def. Given two positive angles α and β,

1. α and β are **complementary** if $\alpha + \beta = \frac{\pi}{2} = 90^\circ$

2. α and β are **supplementary** if $\alpha + \beta = \pi = 180^\circ$

complementary angle

supplementary angles



ex. Find the complement and supplement of the angle $\theta = \frac{\pi}{7}$.

complement

$\beta + \frac{\pi}{7} = \frac{\pi}{2}$ $\qquad \beta = \frac{\pi}{2} - \frac{\pi}{7} = \frac{7\pi}{14} - \frac{2\pi}{14} = \frac{5\pi}{14}$

supplementary angle

$\beta + \frac{\pi}{7} = \pi$ $\qquad \beta = \pi - \frac{\pi}{7} = \frac{7\pi}{7} - \frac{\pi}{7} = \boxed{\frac{6\pi}{7}}$

convert to radians

$$30^\circ\left(\frac{\pi}{180^\circ}\right) = \frac{\pi}{6}$$

$$1 = \frac{180^\circ}{6}$$

Degree Measure

Another way to measure angles is in terms of **degrees**, denoted by °.

$$1 \text{ counterclockwise revolution} = 360^\circ$$

$(180^\circ = \pi \text{ rad})\frac{1}{180}$

NOTE: $1 \text{ revolution} = \boxed{360^\circ = 2\pi \text{ rad}}\ \frac{1}{2}$

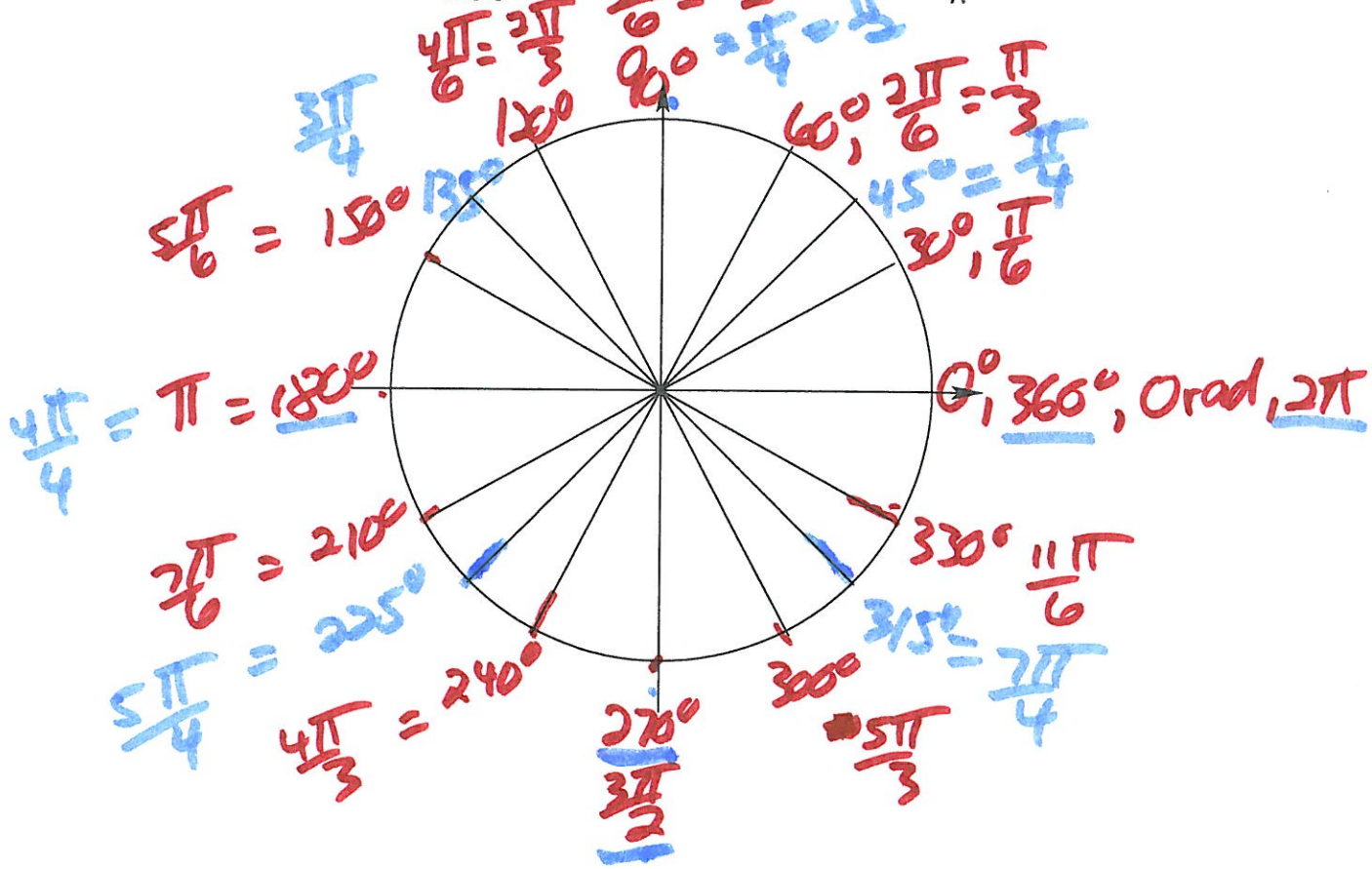
$$\Longrightarrow \boxed{180^\circ = \pi \text{ rad}}$$

$* \ \frac{1}{\pi}(180^\circ = \pi \text{ rad})$

Therefore, we have

$$1^\circ = \frac{\pi}{180} \text{ rad} \quad \text{and} \quad 1 \text{ rad} = \frac{180^\circ}{\pi}$$

$\frac{180^\circ}{\pi} = 1 \text{ rad}$



Conversions between Radians and Degrees

$$\text{Radian} \underset{\times \frac{\pi}{180^\circ} = 1}{\overset{\times \frac{180^\circ}{\pi} = 1}{\rightleftarrows}} \text{Degree}$$

ex. Convert each angle in degrees to radians.

1) $\cancel{60}^{1}{}^\circ \times \frac{\pi}{\cancel{180}_{3}} = \frac{\pi}{3}$ radians

2) $150^\circ \cdot \frac{\pi}{180} = \frac{5\pi}{6}$ rad.

ex. Convert each angle in radians to degrees.

1) $\frac{\pi}{6} \cdot \frac{180^\circ}{\pi} = 30^\circ$

2) $-\frac{3\pi}{4} \cdot \frac{180^\circ}{\pi} = -135^\circ$

p.3 $\theta = \frac{s}{r}$ ← arc length

↖ radius

Solve for s

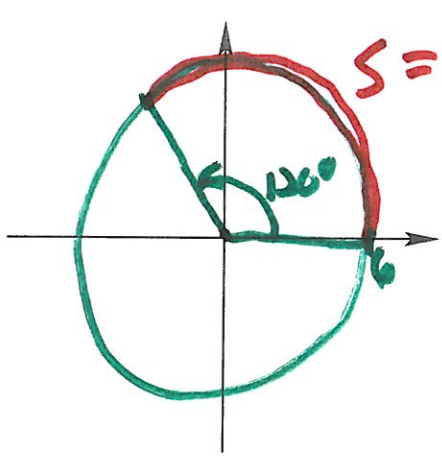
Arc Length

For a circle of radius r, a central angle θ intercepts an arc of length s given by

$$s = r\theta$$

where θ is measured in radians.

ex. A circle has a radius of 6 inches. Find the length of the arc intercepted by a central angle of 120°.



s = ? inches

$s = r\theta$

$s = 6\left(\frac{2\pi}{3}\right) = 4\pi$ radians

$\approx 4(3.14) \approx 12.56$ in.

$\cancel{120}^{2}\left(\frac{\pi}{\cancel{120}_{3}}\right) = \frac{2\pi}{3}$

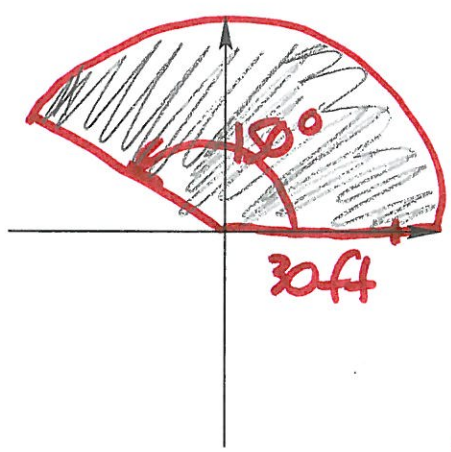
Area of a Sector

For a circle of radius r, the area A of a sector with central angle θ is given by

$$A = \frac{1}{2}r^2\theta$$

where θ is measured in radians.

ex. A sprinkler sprays water over a distance of 30 feet while rotating through an angle of 150°. What area of lawn receives water?



$A = \frac{1}{2}(30^2)\frac{5\pi}{6} = \boxed{375\pi \text{ square feet}}$

$\approx 375(3.14) = 1176.88$ ft

$150° \cdot \frac{\pi}{180°} = \frac{5\pi}{6}$

1 ft

$\boxed{A = 1\cancel{2}\text{ ft}}$ 1ft

distance = rate × time

rate = $\frac{\text{distance}}{\text{time}}$

Linear and Angular Speed

Consider an object moving at constant speed along a circular arc of radius r.

1. If s is the length of the arc traveled in time t, then the **linear speed** v of the object is

$$v = \frac{s}{t}$$

arc length

linear speed

time

omega

2. If θ is the angle (in radian mesure) corresponding to the arc length s, then the **angular speed** ω of the object is

how many degrees or radians are gone through

$$\omega = \frac{\theta}{t}$$

angular speed

NOTE:

speed moving along the arc

1. Linear speed measures how fast the object is moving, and angular speed measures how fast the angle is changing.

2. The relationship between linear speed v and angular speed ω:

$$v = r\omega$$

p.3 $\theta = \frac{s}{r}$

p.8 $s = r\theta$

p.10 $v = \frac{s}{t} = \frac{r\theta}{t} = r \cdot \frac{\theta}{t} = r\omega$ ths $v = r\omega$

ex. A circular gear in a motor rotates at the rate of 100 rpm (revolutions per minute).

1) What is the angular speed of the gear in radians per minute? $100(2\pi \text{ radians})$ thus $\frac{200\pi \text{ radians}}{\text{minute}}$

or $\omega = \frac{\theta}{t} = \frac{100(2\pi)}{1 \text{ min}}$

2) What is the linear speed of the gear in cm/sec if the radius of the gear is 4 cm? $r = 4$

linear speed: $v = \frac{s}{t}$

per minute

arc length: $s = r\theta$, so $s = (4\text{cm})(200\pi)$

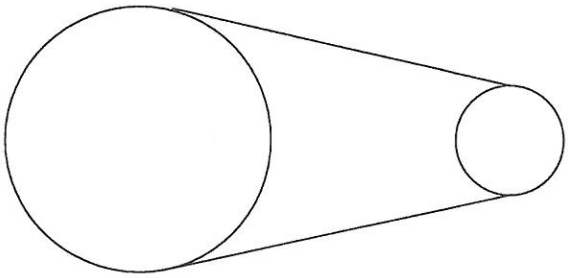
so $v = \frac{(4\text{cm})(200\pi)}{1 \text{ min}} = \frac{(\cancel{4}\text{cm})(\overset{40}{\cancel{200}}\pi)}{\underset{\frac{\cancel{45}}{3}}{\cancel{60}} \text{ sec}} = \boxed{\frac{40\pi}{3} \text{ cm/sec}}$

convert to seconds

can use $d = r \cdot t$

Practice.

1) Two pulleys, one with radius 2 inches and the other with radius 8 inches, are connected by a belt. If the 2-inch pulley is caused to rotate at 6 rpm, determine the revolutions per minute of the 8-inch pulley.



2) Find the angle of least postive measure that is coterminal with $\theta = -695°$.

Answer. 1) $\frac{3}{2}$ rev/min 2) $25°$